SCCCI × NYP AI INNOVATION CHALLENGE

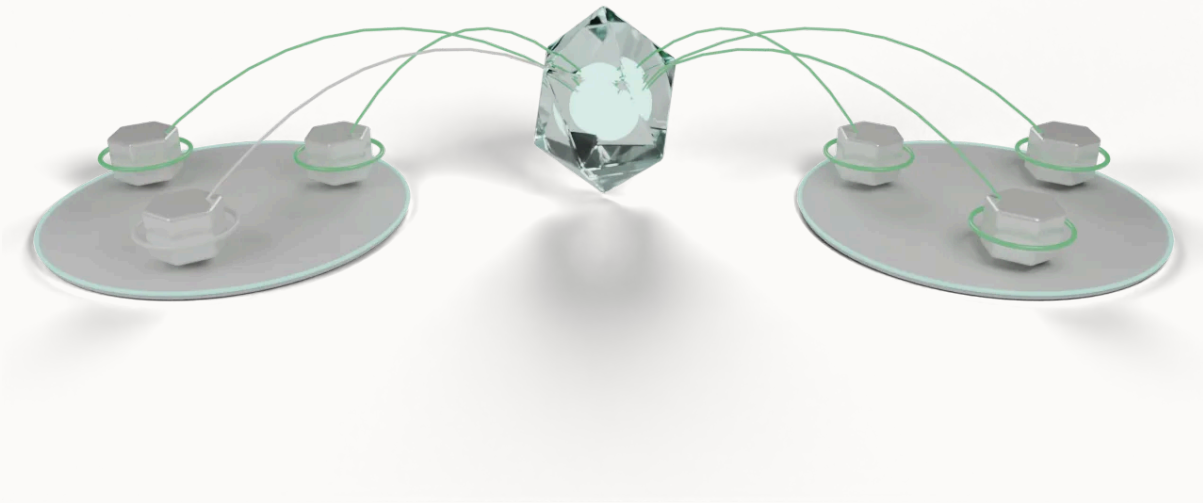
HuiTong 汇通

An integration and workflow layer for Organisation E's HR systems

Richie · Iok Teck · Azri · Angelyn

Problem Statement 17 · IT2134 IT Innovation Project

Nanyang Polytechnic · AY2026 Semester 2



THE PROBLEM

Mei Ling is the integration

A Singapore HR executive, on the second working day of the month

08:45
Copies an approved OA request into her recruitment tracker

09:30
Keys yesterday's joiner into SG HR, then into the Excel master

11:00
Works out which job title is right when SAP and SG HR disagree

13:30
Exports attendance and leave, and asks HQ for theirs on WeChat

15:30
Fixes HQ's garbled Chinese headers and YYYY/MM/DD dates

18:30
Leaves with the month's consolidation half done

> "We have four systems, and I am still the one connecting them, one spreadsheet at a time."

A proto-persona with an illustrative quote, built from the problem statement and desk research, to be validated with Organisation E.

ROOT CAUSE

Digital, but isolated

Every system works on its own. Nothing connects them, so people do.



From Problem Statement 17: approvals happen in OA but the follow-up work happens elsewhere, and the Excel master is the only consolidated view.

WHAT SUCCESS LOOKS LIKE

Enter it once, and let it flow

~16 h → **≤ 2 h**

HR hours a month consolidating attendance and leave

3–5 work days → **< 5 min**

From an OA approval to the onboarding checklist

~3 → **1**

Times each new hire's details are typed by hand

Weeks → **< 1 day**

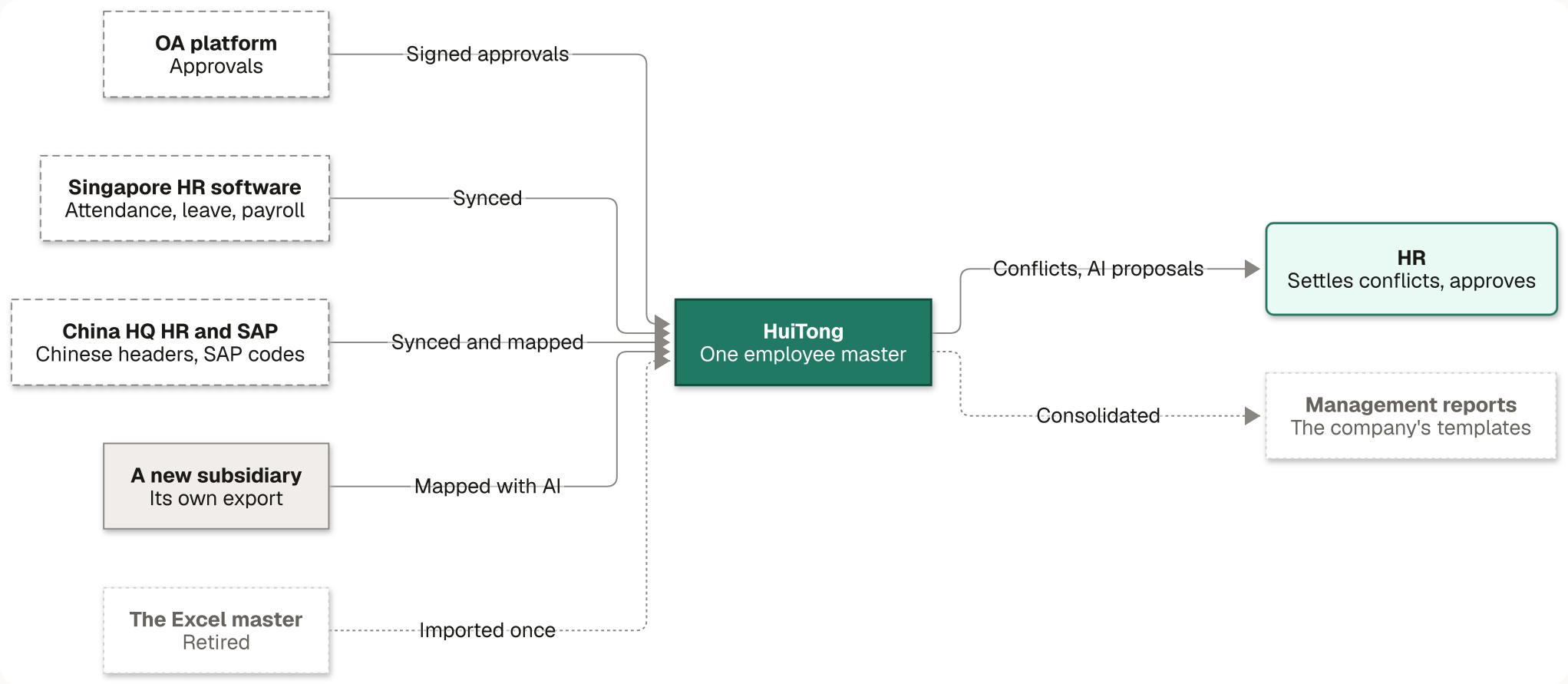
To connect a new subsidiary's HR export

The baselines are assumptions until we measure them with Organisation E. No gain may cost a guardrail: AI never writes without a person's approval, no national ID numbers are stored, and another entity's data is seen only as the transfer register allows.

THE SOLUTION

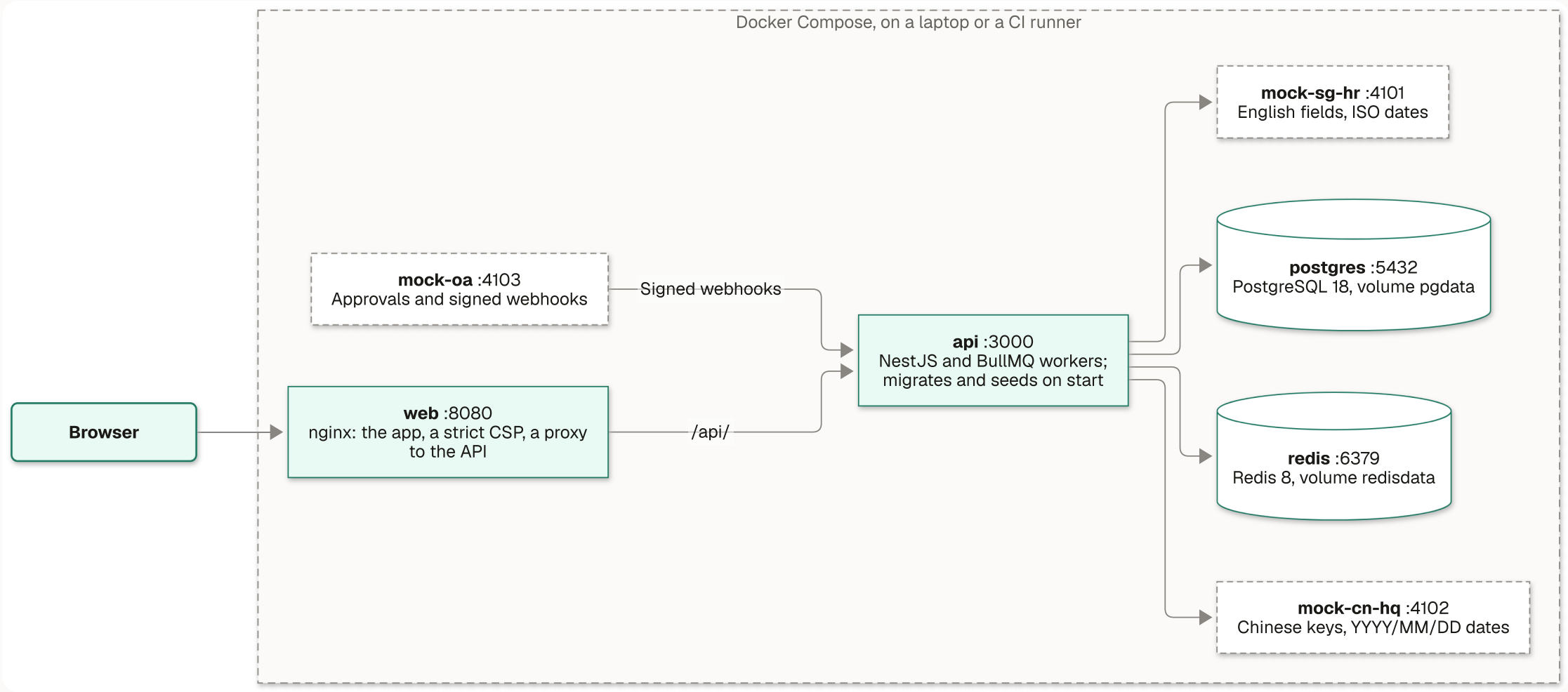
An integration layer, not a replacement

Every system of record stays, behind one adapter each; only the Excel master retires



ARCHITECTURE

One web app, one API, the systems around it



A KEY WORKFLOW

One approval, four areas

Arjun's manpower request for a relationship manager, end to end

1

OA's signed approval arrives, and is stored once

Areas 1 and 2 · Receiving built

2

The requisition, the onboarding checklist and the preparation tasks appear

Area 2

3

A draft employee record, linked to the request

Areas 1 and 2

4

SG HR's verified record activates the draft

Area 1 · Sync built

5

Reconciled with HQ's record by the group employee number

Area 1 · Matching built

6

Later leave imported and mapped: 年假, 病假, 调休

Area 3

7

HQ's report holds only the fields HQ may hold

Areas 3 and 4 · Area 1's allowlist built

8

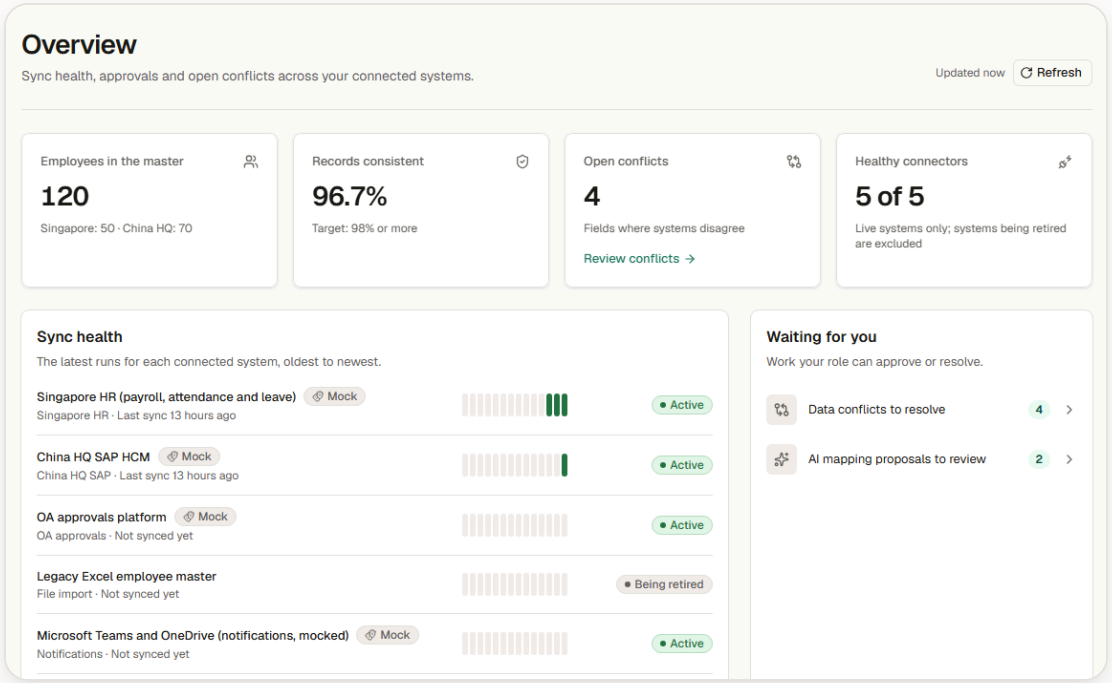
The overview shows the history and any exceptions

All areas · Overview built

Playwright runs this journey as E2E #1 on every pull request: step 1 runs today, and each other step is marked with its story until that story lands.

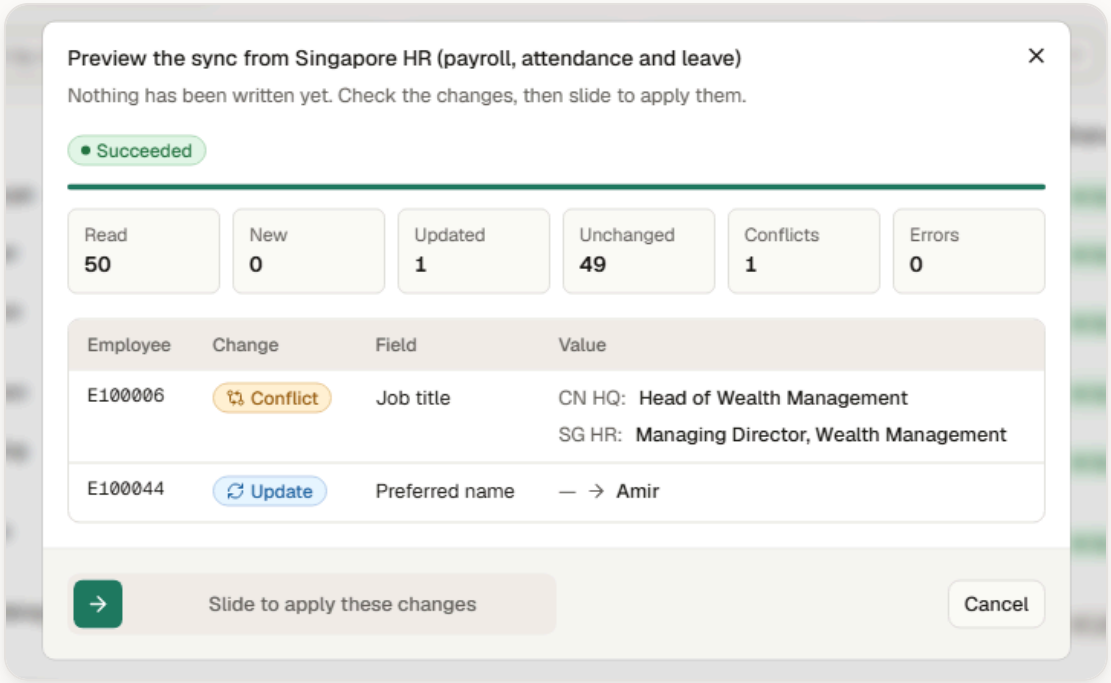
THE PRODUCT TODAY

Built on the real API, with synthetic data



The overview

Sync health and the work waiting for HR; every mock is labelled



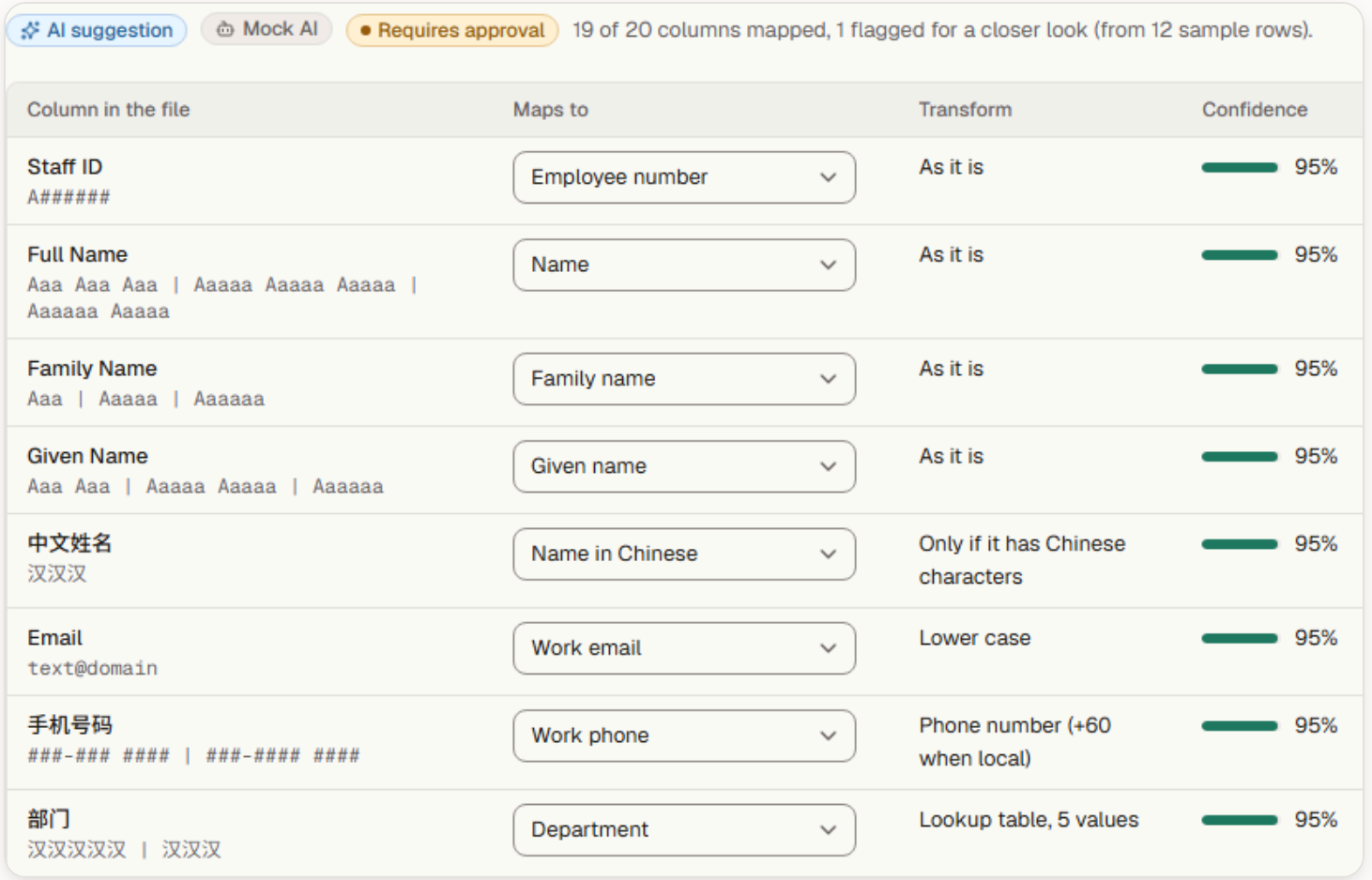
A sync, previewed

Nothing is written until HR has seen the changes and the conflicts

AI, WITH A PERSON IN CHARGE

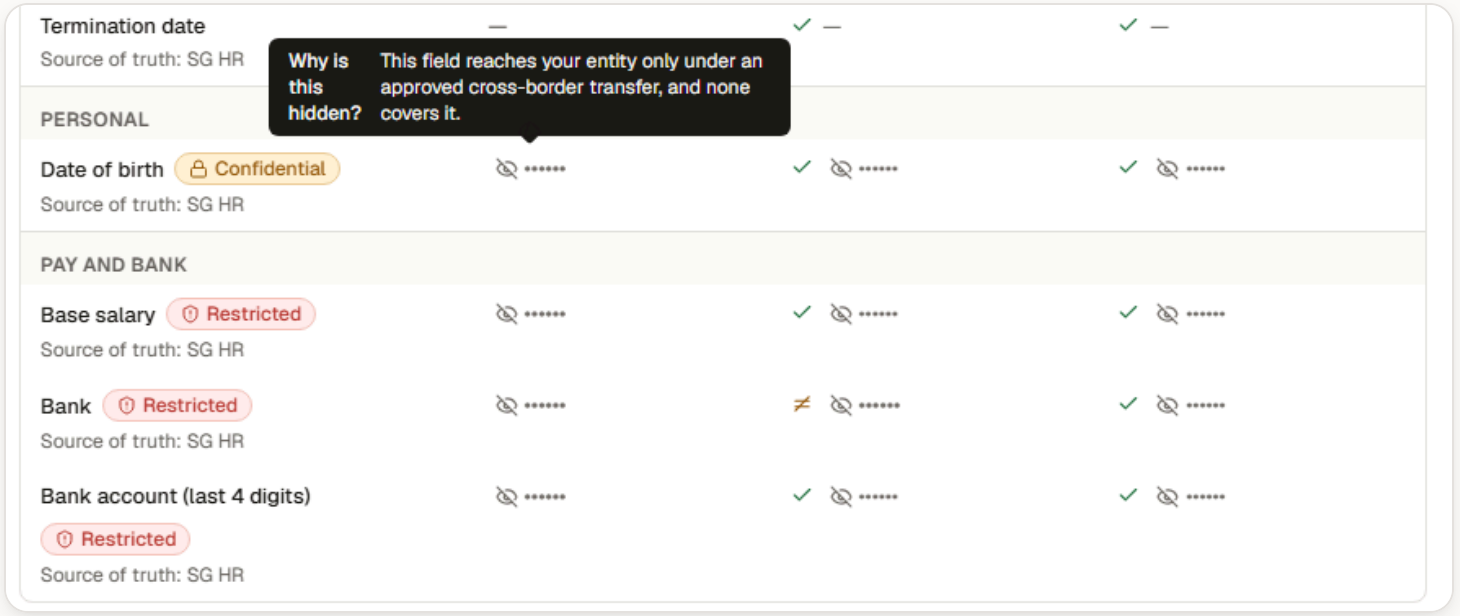
The AI proposes; HR decides

- **Maps** columns to the master, Chinese headers too
- **Explains** disagreements, in English or 中文
- **HR decides,** and every approval is audited
- **No identifiers** ever reach the model
- **Every call** recorded: tokens, cost, outcome



Controls built in, not bolted on

- **Least privilege:** seven roles, by entity
- **Masked fields** say why they are hidden
- **Preview first:** HR sees a sync before it writes
- **Every write audited,** in a hash chain
- **Cross-border:** default deny, by register



Seen by HQ HR: the master, CN HQ and SG HR columns, with each hidden field's reason

DEPLOY AND MONITOR

Watching it run: health, usage and AI cost

Observability
Requests, errors and the error rate

Requests	Errors	Error rate
94.7K	296	0.31%

Clickstream
First-party, from consenting sessions only

Sessions	Page views	Views per session
442	2,910	6.6

AI usage
Calls, tokens, estimated cost and latency

Calls	Tokens	Estimated cost	Error rate	Median latency
533	760.3K 585.3K in, 175K out	$4.38	3.2% Failed calls and output that failed validation. A proposal HR rejected is not an error.	50ms

The dashboards show labelled sample data until area 4 builds the aggregations, in weeks 14 to 16.

NEXT

Roadmap, and what we need to ask

W5
The proposal, on this prototype

W8 to W12
Each area builds its core logic

W13
The technical review

W14 to W16
Observability, clickstream and AI token tracking

W18 and W19
The final presentation, then the client check-in

Questions for Organisation E

- Which HR products run in Singapore and at HQ, and what APIs or exports do they offer?
- Which OA product runs the approvals?
- Is there one group employee number in both systems?
- Where may employee data be stored and processed?
- May HuiTong write back to the systems of record?

THE TEAM

Four areas, each owned end to end

AREA 1

Integration core and platform

Connectors, the employee master, mappings and the AI assistant, syncs and conflicts, access enforcement, the platform and the integration map

Richie, team lead

AREA 2

Workflow automation

OA approvals through to onboarding: the state machine, retries, SLAs and notifications

Owner to be named by W3

AREA 3

Attendance, leave and reporting

Both HR systems' attendance and leave, the category mapping, the company template and reports

Owner to be named by W3

AREA 4

Governance, security and monitoring

Administering access and the transfer register, the audit-log viewer, observability, clickstream and AI cost, and the AWS deployment

Owner to be named by W3

Richie, Iok Teck, Azri and Angelyn. From week 8 each member builds the area they own, and explains that code at W13 and W18.